Second: Homework Task – Model Answers P.9

1- Complete the following sentences:

a. Air enters the body through the nose and mouth.

b. The lungs are the site where gases are exchanged in the body.

c. Blood transports oxygen gas to the whole body.

d. Fish use gills for breathing underwater.

e. Oxygen moves into the blood through the capillaries around the alveoli in the lungs, and then the heart pumps it to the whole body.

2- Give reason for each of the following:

a. Gas exchange takes place in the alveoli.

Because they have blood capillaries that allow blood to move into the blood and carbon dioxide move from the blood into the alveoli to be exhaled

b. Limewater turns milky when exhaled air is passed through it.

Because exhaled air contains carbon dioxide, which reacts with limewater and makes it milky.

c. Living organisms need oxygen.

Because oxygen is needed for respiration to release energy from food.

d. Fish have gills.

Because gills absorb dissolved oxygen from water.

3- Compare between each of the following:

a. Inhalation and exhalation in terms of the type of gas and the direction of air movement.

Inhalation: oxygen-rich air enters the lungs. Exhalation: carbon-dioxide-rich air leaves the lungs.

b. Breathing in fish and breathing in cats in terms of the respiratory organ.

Fish breathe using gills; cats breathe using lungs.

4- Answer the following questions:

a- Why is it important to keep the air in the room fresh?

To provide fresh oxygen and remove excess carbon dioxide and other pollutants.

b- Why can't fish live out of water for a long time?

Because their gills need water to get dissolved oxygen and usually cannot function properly in air.

c- Describe the path of oxygen and name the organs through which it passes until it reaches the blood.

Oxygen enters through the nose or mouth, passes through the trachea and bronchi into the lungs, reaches the alveoli, and diffuses into the blood in surrounding capillaries.

d- What happens to the body if it does not get enough oxygen?

Cells release less energy, causing weakness.

f- Why does limewater turn milky when exhaled air is passed through it?

Because carbon dioxide in exhaled air reacts with limewater to form calcium carbonate, which makes it look milky.